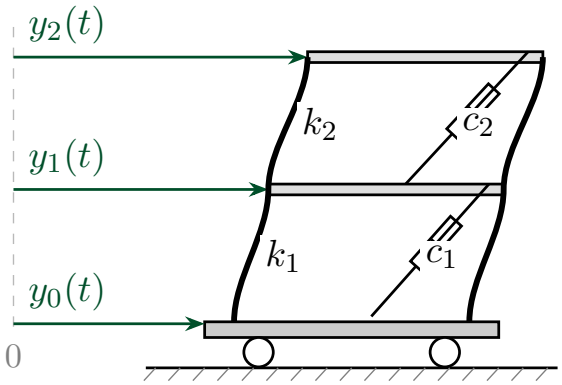
MA3521 — Building Models and Linear ODEs — Key

Name:

1. **Two-Story Building Practice**

Displacements: $\mathbf{y}(t) = (y_1(t), y_2(t))$ (m), t in seconds.



$m_2 = 2\,\text{kg}$
$k_2 = 4\,\text{N/m}$
$c_2 = 2\,\text{N s/m}$

$m_1 = 3\,\text{kg}$
$k_1 = 6\,\text{N/m}$
$c_1 = 3\,\text{N s/m}$

$y_0(t) = 2\sin(t)$ m

Displacements are in m and time is in s. The matrices have units kg, N s/m, and N/m, respectively, and $\mathbf{b}$ is in N.

(a) For $\boldsymbol{M}\mathbf{y}'' + \boldsymbol{C}\mathbf{y}' + \boldsymbol{K}\mathbf{y} = \mathbf{b}(t)$

$$\boldsymbol{M} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \qquad \boldsymbol{C} = \begin{pmatrix} 5 & -2 \\ -2 & 2 \end{pmatrix} \qquad \boldsymbol{K} = \begin{pmatrix} 10 & -4 \\ -4 & 4 \end{pmatrix}$$

$$\mathbf{b}(t) = \begin{pmatrix} 6y_0(t) + 3y_0'(t) \\ 0 \end{pmatrix}$$

The base spring and damper act on the first floor. With $y_0(t) = 2\sin(t)$ this becomes

$$\mathbf{b}(t) = \begin{pmatrix} 12\sin(t) + 6\cos(t) \\ 0 \end{pmatrix}$$

(b) With $\mathbf{z} = (y_1, y_2, y_1', y_2')$

$$\boldsymbol{A} = \begin{pmatrix} \mathbf{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix}$$

For completeness, the forcing in $\mathbf{z}' = \boldsymbol{A}\mathbf{z} + \mathbf{g}(t)$ is

$$\mathbf{g}(t) = \begin{pmatrix} 0 \\ 0 \\ 2y_0(t) + y_0'(t) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 4\sin(t) + 2\cos(t) \\ 0 \end{pmatrix}$$

2. **Linear Initial Value Problem**

$$y' - \frac{4}{t}y = 2t^5 \qquad y(1) = 3$$

(a) The integrating factor is

$$\mu(t) = \exp\left(\int \boxed{-4/t}\, dt\right) = \boxed{t^{-4}}$$

since $\int(-4/t)\,dt = -4\ln|t|$.

(b) Multiplying by $\mu(t)$ gives

$$(t^{-4}y)' = \boxed{2t}$$

$$t^{-4}y = \int \boxed{2t}\, dt + C = \boxed{t^2} + C$$

(c) The initial condition gives $3 = 1 + C$, so $C = 2$ and

$$\boxed{t^{-4}y = t^2 + 2}$$

Check

$$(t^{-4}y)' = 2t \qquad 1^{-4}y(1) = 1^2 + 2 = 3$$